Xinran Zhang

EDUCATION BACKGROUND

Tsinghua University, Beijing, China Sept. 2024 – Jun. 2027 (expected)
- Master of Mechanical Engineering | Mechanical Engineering: A+ in MOE National Discipline Assessment
- GPA: 4.0 / 4.0 (Rank: **1/74**)
- Honors: University First-Class Scholarship (2025)

Hunan University, Changsha, China Sept. 2020 – Jun. 2024
- Bachelor of Vehicle Engineering | Mechanical Engineering: A- in MOE National Discipline Assessment
- GPA: 3.91 / 4.0 (Rank: **1/123** for three consecutive academic years)
- Honors: National Scholarship (2021, 2022, 2023), Outstanding Graduate, Exemplary Student Award

RESEARCH INTERESTS

Physical signal-based model for industrial and robotic systems: multimodal basic model for physical signals such as vibration, acoustics, and haptics; physical-signal-based large models in robot tactile perception / embodied intelligence; industrial intelligent agent.

PUBLICATIONS AND MANUSCRIPTS

[1] **Xinran Zhang**, Jinfeng Huang, et al. VibAlign: Learning Language-Aligned Signal Tokens for Vibration Signal Analysis with Large Language Models. *Submitted to Mechanical Systems and Signal Processing.*

[2] **Xinran Zhang**, Qi Li, et al. IADG-BiMamba: Imbalance-Aware Domain Generalization with Bidirectional Mamba for Cross-Condition Bearing Fault Diagnosis. *Submitted to IEEE Sensors Journal.*

[3] **Xinran Zhang**, Jinfeng Huang, et al. FaultOvis: A Domain-Generalized Vision-Language Model for Rolling Bearing Fault Diagnosis. *Submitted to IEEE Transactions on Instrumentation And Measurement.*

[4] Qi Li†, **Xinran Zhang**†, et al. VSLLaVA: A pipeline for Large Multimodal Foundation Models in industrial vibration signal analysis. *Advanced Engineering Informatics*, 2026, 76: 105023.

†Equal contribution.

RESEARCH EXPERIENCE

Physical Signal Foundation Models for Industrial and Robotic Systems Nov. 2025 – Present
Master's Thesis Project, Tsinghua University
- Investigate how LLMs and MLLMs can be adapted to non-text physical signals, using vibration and robotic tactile data as testbeds for sensor-based perception, cross-modal alignment, and natural-language reasoning.
- Advance a progressive research pipeline from signal-image-based visual instruction tuning to domain-generalized VLM representation learning and raw 1-D signal tokenization, leading to three paper-level subprojects on physical-signal foundation model adaptation.

Raw Physical Signal Tokenization and Language Alignment for LLM Jun. 2026 – Present
Graduate Research Project, Tsinghua University
- Developed VibAlign, a raw physical-signal-to-LLM framework that maps 1-D vibration segments from industrial and robotic tactile sensing scenarios into continuous LLM-compatible signal-token embeddings.
- Designed a BiMamba-based temporal signal encoder, a learnable soft codebook connector, and a three-stage signal-language alignment pipeline combining encoder pretraining, signal-text contrastive alignment, and LoRA-based LLM adaptation with verbalizer-guided supervision.
- Evaluated VibAlign on robotic tactile and industrial vibration benchmarks, achieving **91.84%/91.34%** Acc./F1 on tactile material recognition and **98.18%/98.17%** average Acc./F1 on public industrial vibration benchmarks, with further validations on candidate scoring, signal-embedding utilization, ablations, and cross-dataset generalization.

Domain-Generalized VLM Adaptation for Industrial Sensor Images Oct. 2025 – Apr. 2026
Graduate Research Project, Tsinghua University

- Developed FaultOvis, an Ovis2-1B-based vision-language adaptation framework that uses STFT images as physical-signal visual inputs for domain-shifted industrial signal reasoning.
- Introduced visual-side discriminative supervision with an auxiliary classification head, a contrastive learning head, and a two-stage training strategy to improve category separability and domain-invariant visual representation learning.
- Evaluated FaultOvis on CWRU and Ottawa leave-one-domain-out benchmarks, achieving the highest Acc./F1 on 7 of 8 unseen-domain tasks and up to **99.33%/99.00%** Acc./F1, with F1 improvements of up to **+76.05** percentage points over the base Ovis2-1B.

Instruction Tuning and RL Refinement for Physical-Signal LMMs Apr. 2025 – Sep. 2025
Graduate Research Project, Tsinghua University

- Developed VSLLaVA, a large multimodal model adaptation pipeline that formulates physical signal analysis as visual question answering for signal type identification, parameter reasoning, and explanatory responses.
- Constructed an expert knowledge-guided Signal-Question-Answer dataset from simulated and real vibration signals, encoding signal characteristics, physical parameters, frequency-domain patterns, and domain knowledge into multimodal instruction-tuning data.
- Fine-tuned six VLM backbones with LoRA-based SQA-SFT, improving Ovis2-8B Word Recall from 64.29% to 80.81% and CIDEr from 0.16 to 5.52; GRPO further improved InternVL3-8B signal-type identification Acc./F1 from 79.36%/76.99% to 97.50%/97.48%.

COMPETITION AWARDS

Honorable Mention, Mathematical Contest in Modeling (MCM/ICM)	2023
National First Prize, National English Competition for College Students (NECCS)	2023
National Third Prize, The 14th Chinese Mathematics Competition for College Students	2023
Second Prize, Hunan Provincial Mechanics Competition	2022

TEACHING & LEADERSHIP EXPERIENCES

Teaching Assistant 2024 – 2025
Tsinghua University

- Supported *Fundamentals of Engineering Drawing* through SolidWorks labs, grading, Q&A, and proctoring for undergraduates; rated **Excellent** in the final TA evaluation.

Leader, Peer Learning Support Group 2021 – 2024
Hunan University

- Led peer study and exam-preparation activities for classmates; received the **Star of Academic Support** award.

TECHNICAL SKILLS AND LANGUAGE

Programming	Python, MATLAB, C/C++, LaTeX
Deep Learning	PyTorch, Hugging Face Transformers, LLM/VLM adaptation, LoRA/PEFT, multimodal instruction tuning, GRPO/RL refinement, contrastive learning, representation learning
Signal Processing	Vibration and tactile signal preprocessing, time-frequency analysis, short-time Fourier transform, envelope spectrum analysis, feature extraction and visualization
Tools	Linux, Git, VS Code, Origin, Visio, AutoCAD, SolidWorks
English	TOEFL iBT 5.5/6.0, equivalent to 110 on the previous 0–120 scale